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Roll no- 08

Practical no- 9

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import os
os.environ["OMP_NUM_THREADS"] = "1"
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans, AgglomerativeClustering
from sklearn.preprocessing import StandardScaler
from scipy.cluster.hierarchy import dendrogram, linkage
import numpy as np
# Load dataset
X = load_iris().data
X = StandardScaler().fit_transform(X)

# Elbow Method
wcss = []
for k in range(1, 11):
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    km.fit(X)
    wcss.append(km.inertia_)

plt.plot(range(1, 11), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()

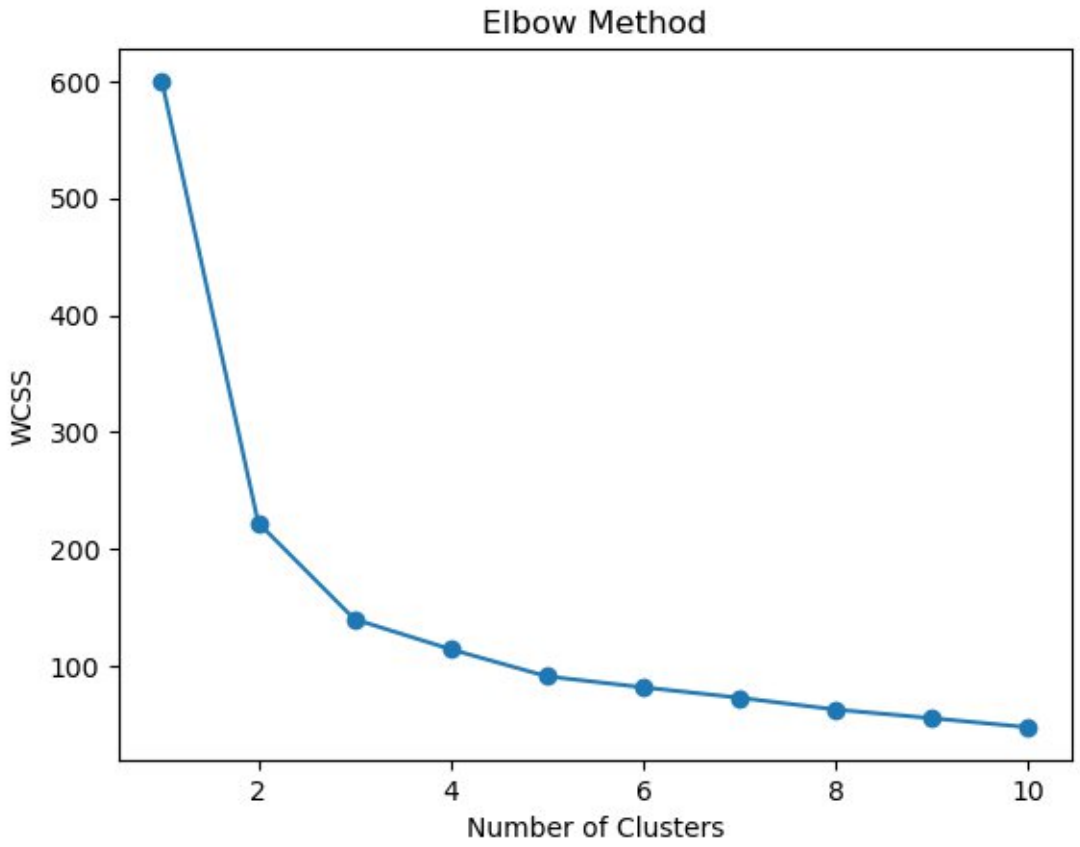
# K-Means Clustering
km = KMeans(n_clusters=3, random_state=42, n_init=10)
```

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kmeans_labels = km.fit_predict(X)
print("K-Means Cluster Sizes:", np.bincount(kmeans_labels))

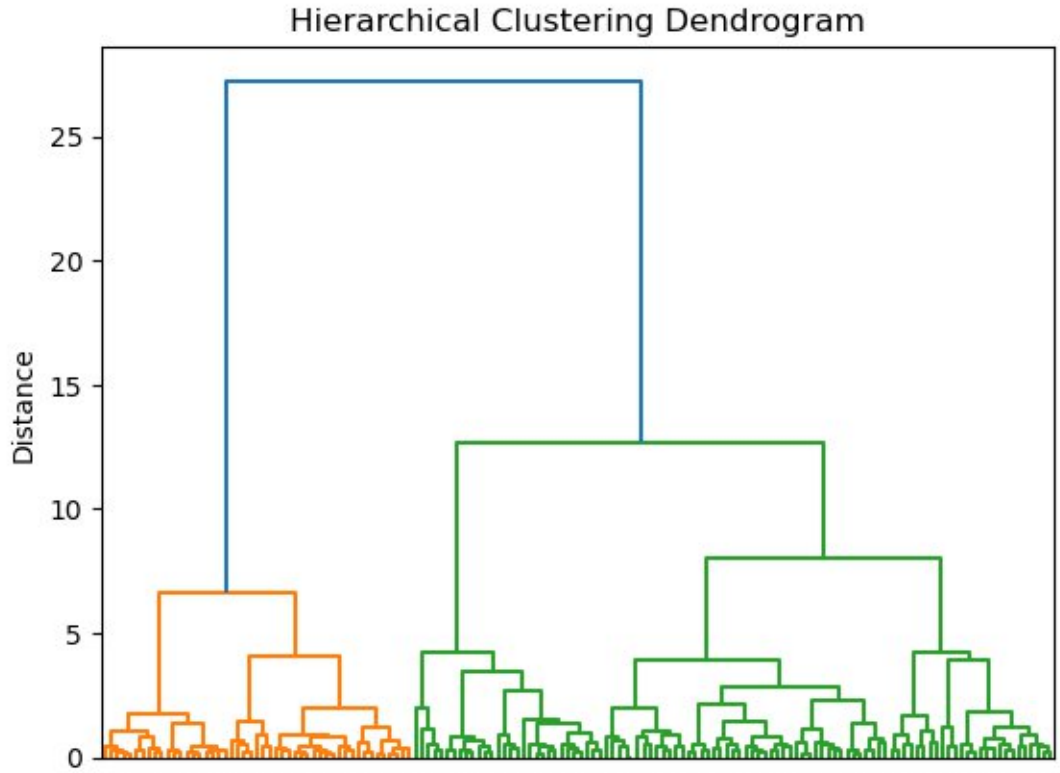
# Dendrogram
Z = linkage(X, method='ward')
dendrogram(Z)
plt.title("Hierarchical Clustering Dendrogram")
plt.xlabel("Samples")
plt.ylabel("Distance")
plt.show()

# Hierarchical Clustering
hc = AgglomerativeClustering(n_clusters=3)
hc_labels = hc.fit_predict(X)
print("Hierarchical Cluster Sizes:", np.bincount(hc_labels))
```

OUTPUT-



K-Means Cluster Sizes: [53 50 47]



Hierarchical Cluster Sizes: [71 49 30]